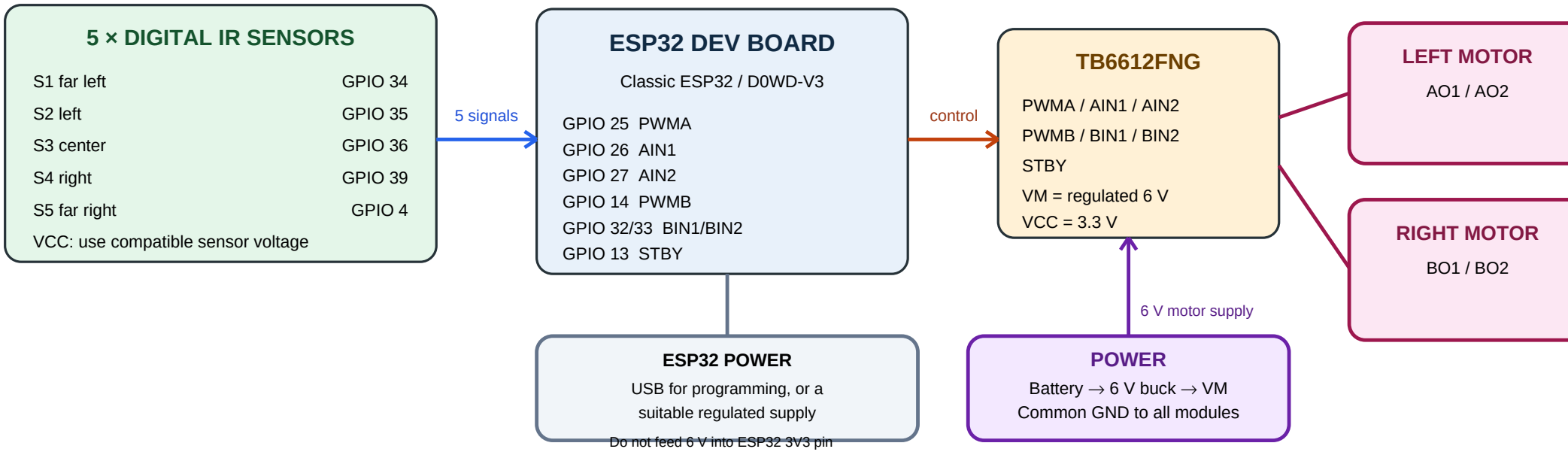
ESP32 5-Sensor Line Follower — Circuit Diagram & Wiring

Based on the pin assignments used in the tuning firmware: classic ESP32 (ESP32-D0WD-V3), TB6612FNG dual motor driver, five digital IR sensors, and two DC drive motors.



A. Pin-to-pin wiring

Module / signal	Connect to	Notes
TB6612FNG PWMA	ESP32 GPIO 25	Left motor speed PWM
TB6612FNG AIN1	ESP32 GPIO 26	Left motor direction
TB6612FNG AIN2	ESP32 GPIO 27	Left motor direction
TB6612FNG PWMB	ESP32 GPIO 14	Right motor speed PWM
TB6612FNG BIN1	ESP32 GPIO 32	Right motor direction
TB6612FNG BIN2	ESP32 GPIO 33	Right motor direction
TB6612FNG STBY	ESP32 GPIO 13	Must be HIGH for driver enabled
TB6612FNG VCC	ESP32 3.3 V	Logic supply
TB6612FNG VM	Regulated 6.0 V	Motor supply; not raw 2S/3S battery
TB6612FNG GND	Common GND	Join ESP32, driver, battery/buck grounds
TB6612FNG AO1/AO2	Left motor terminals	Swap wires if forward direction is reversed
TB6612FNG BO1/BO2	Right motor terminals	Swap wires if forward direction is reversed

Module / signal	Connect to	Notes
IR sensor S1 output	ESP32 GPIO 34	Far left
IR sensor S2 output	ESP32 GPIO 35	Left
IR sensor S3 output	ESP32 GPIO 36	Center
IR sensor S4 output	ESP32 GPIO 39	Right
IR sensor S5 output	ESP32 GPIO 4	Far right
IR sensor VCC / GND	Compatible supply / common GND	Verify module voltage and output levels

B. Power and safety notes

1. Motor voltage: The N20 motors discussed for this build are rated for 6 V. Feed TB6612FNG VM from a buck regulator adjusted to 6.0 V. Do not connect a 2S or 3S LiPo directly to VM when using 6 V motors.

2. ESP32 supply: Keep the ESP32 powered by USB while uploading/debugging, or use the board's approved regulated input. Never connect 6 V to the ESP32 3V3 pin.

3. Common ground: Connect the ESP32 GND, TB6612FNG GND, sensor GND, and buck-regulator output GND together.

4. IR output voltage: ESP32 GPIOs are not 5 V tolerant. If the sensor board outputs 5 V logic, use a level shifter or a suitable voltage divider before connecting its signal to the ESP32. Use 3.3 V sensor power only if the sensor module supports it.

5. GPIO 34–39: On the classic ESP32, these pins are input-only. That is appropriate for the IR sensor outputs; do not use them for motor outputs.

6. First test: Raise the wheels off the ground before pressing RUN. Verify left/right motor directions and sensor states at low speed. Press STOP if the robot turns the wrong way.

7. Motor current: Make sure your buck regulator and wiring can supply the motors' combined current. The TB6612FNG has current limits; avoid prolonged motor stalls.

8. Diagram scope: This is a functional wiring diagram, not a physical PCB layout. Check the labels printed on your specific TB6612FNG and sensor modules before powering them.

C. Firmware pin map

The diagram matches the code pin definitions: PWMA=25, AIN1=26, AIN2=27, PWMB=14, BIN1=32, BIN2=33, STBY=13; sensors S1=34, S2=35, S3=36, S4=39, S5=4. The default firmware assumes the sensor output is LOW when the line is detected; the web dashboard includes a HIGH/LOW option.